

Indium mineralization of the Chaganbulagen Ag–Pb–Zn deposit in Inner Mongolia, Northeast China: Evidence from geochemistry of sphalerite and chalcopyrite[☆]

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ABSTRACT

The Chaganbulagen large Ag–Pb–Zn deposit in the southwestern segment of the Derbugan metallogenic belt contains significant indium (In) and cadmium (Cd), though their enrichment mechanisms remain unclear. Field geological investigations and petrographic observations revealed well-developed mineralization zoning. Furthermore, four mineralization stages were identified, comprising four generations of sphalerite (SpI–SpIV) and chalcopyrite (CcpI–CcpIV). LA-ICP-MS analysis shows In and Cd mainly occur in sphalerite via isomorphism, followed by chalcopyrite. Cadmium exhibits consistent enrichment across all mineralization stages, whereas indium displays a marked decline from early-stage (SpI: averaging 656 ppm; CcpI: averaging 270 ppm) to late-stage sulfides (SpIV: averaging 1.15 ppm; CcpIV: averaging 0.36 ppm). This depletion trend correlates with decreasing temperatures (336–208 °C) and diminishing Cu–Fe concentrations in the ore-forming fluids. Spatially, shows two trends in indium distribution: (1) vertical increasing with depth in No. X orebody, and (2) lateral zoning with decreasing In content away from the porphyry Mo (Cu) mineralization, highest in proximal Cu–Zn stage sphalerite (up to 1112 ppm at 1.5–3 km), moderate in intermediate Cu–Pb–Zn and Pb–Zn–Ag stages orebodies (up to 76.8 ppm at 3–5.5 km), and negligible in distal Ag stage orebodies (< 5 ppm at 5.5–8 km). Indium in chalcopyrite follows a similar enrichment pattern as in sphalerite. Based on this, the Jia–Cha ore field, centered on the porphyry body within 3 km to the northwest and southeast, and the deeper parts of the field are favorable areas for future indium ore exploration.

1. Introduction

Indium, as a strategic critical minerals resource, is widely used in modern industrial systems and defense technologies due to its low melting point, superconductivity and optoelectronic properties (Jiang et al., 2019; Zhai et al., 2019). Its strategic significance is closely tied to national technological sovereignty and industrial security. With the acceleration of global digitization and energy transition, indium's role as a core 21st-century strategic resource has become increasingly prominent, and its supply security is now a focal point of competition among major economies (Li et al., 2007, 2019; Chen and Zhao, 2021; Hou et al., 2022).

Indium-rich deposits in China are highly aggregated regionally and

are mostly distributed along block margins (Zhang et al., 2005; Zhao et al., 2022). The south-southwest margin of Yangtze Block is the globally most significant indium resource base (Tu, 2004; Luo et al., 2009; Cheng, 2012; Xu et al., 2021b), represented by Guangxi Dachang (8775 t), Yunnan Dulong (5124 t) and Gejiu (> 4000 t) (Xu and Li, 2018) Sn-polymetallic deposits. Moreover, some indium-rich deposits are also distributed in the Nanling Range region of South China, such as Yejiwei (Liu et al., 2018) and Xianghualing (Ma et al., 2023) Sn-polymetallic deposits, and Qibaoshan Cu-polymetallic deposit (Hu et al., 2017). Significantly, a substantial concentration of large-scale indium-enriched sulfide deposits has been identified within the northern marginal zone of the North China Block, such as Inner Mongolia Meng'entaolegai (> 500 t, Zhang et al., 2006) and Dajing (> 768 t, Ishihara et al., 2008) Cu–Sn

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polymetallic deposits. With the deepening of geological work and increasing interest in indium in recent years, Anle Sn-Cu polymetallic deposit and Weilasituo-Bairendaba Sn—Li—Rb—Cu—Pb—Zn polymetallic deposit in Great Xing'an Range have been further discovered to have indium mineralization and the potential to become medium-to-large-scale indium deposits (Zhu et al., 2021; Huang et al., 2023; Ye et al., 2023), showing the Great Xing'an Range have an excellent geochemical background of indium regionally and is expected to become another important indium-rich deposit gathering zone in China (Zhang et al., 2005; Zhao et al., 2022). However, the occurrence and enrichment mechanism of indium in these deposits are still obscure.

The Chaganbulagen deposit, as a typical large-scale Ag-Pb-Zn deposit in Great Xing'an Range, with the reserves were identified as 289016 t Pb, 297479 t Zn, 1047610 kg Ag, 10891 t Cu, 1511 kg Au, and 2231 t Cd, at an average grade of 2.56 wt% Pb, 2.64 wt% Zn, 164 g/t Ag, 0.14 wt% Cu, 0.14 g/t Au, and 0.022 % Cd. Major metallogenic elements are Pb, Zn, Ag, and associated components are S, Au, Cd and In (XYMCL, 2019). Previous research mainly focus on the genesis of Ag-Pb-Zn polymetallic veins (Zhai et al., 2013; Li et al., 2016; Hui et al., 2021). However, with the emphasis on critical metals, Chaganbulagen deposit were found to be associated with beneficial elements such as In and Cd. The Jia-Cha Pb—Zn—Ag ore field contains considerable In content (780 t) and Cd

(13000 t), with average grades of 304 ppm Cd and 29 ppm In, both reached a large scale (Hui et al., 2021). Currently, an indium resource estimate has been completed for Jiawula deposit, but not for Chaganbulagen deposit. Hui et al. (2021) found indium mineralization in No. X and XI Zn—Pb—Ag orebodies of Chaganbulagen deposit by LA-ICP-MS trace element analysis of sulfide. Thus Chaganbulagen deposit belong to In-Cd-rich deposit, but far less emphasis has been placed on indium. The spatial distribution of Cd and In and their enrichment mechanism is poorly constrained.

This study is based on detailed field geological investigations and indoor petrographic observations of Chaganbulagen deposit. The LA-ICP-MS trace element analysis of sphalerite and chalcopyrite in different mineralization zoning and mineralization stages is used to reveal the occurrence and spatial distribution of indium in Chaganbulagen deposit, and to explore the mechanism of indium enrichment by combining with the significance of characteristic elemental indications of sphalerite. The aim is to provide a theoretical basis for the recovery and utilization of indium resources as well as for prospecting and exploration in Chaganbulagen deposit.

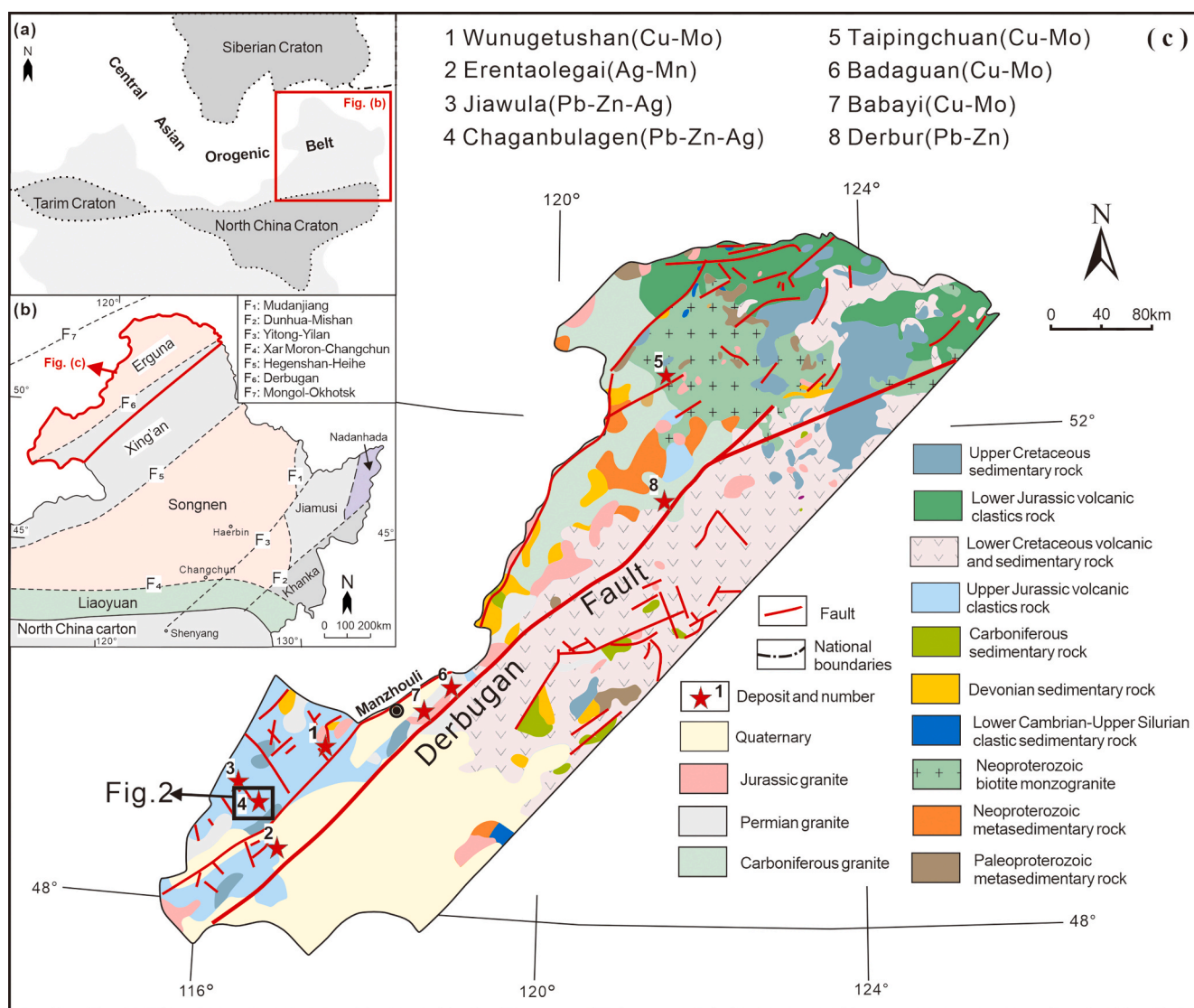


Fig. 1. (a) Schematic map of the geotectonic background of northeast China (modified from Niu et al., 2016). (b) Tectonic map of Northeast China and location of the study area (modified from Wu et al., 2011). (c) Regional Geology and deposits distribution of Derbugan metallogenic belt (modified from Gao et al., 2016).

2. Regional geology

The Derbugan metallogenic belt is located in the Great Xing'an Range region of northeast China, with the geotectonic position in the eastern section of the Central Asian orogenic belt, and between the Siberian Craton and the North China Craton, occupying most of Erguna block (Fig. 1a) (Wu et al., 2011; Niu et al., 2016). A series of NE- and NNE-trending faults divide the Great Xing'an Range into the Erguna, Xing'an, Songnen, Liaoyuan, Jiamusi, and Nadanhada blocks (Fig. 1b). The Erguna, Xing'an, and Songnen blocks merged in the Paleozoic, and the Jiamusi and Nadanhada blocks accreted in the Mesozoic (Wu et al.,

2011). Paleozoic tectonic evolution of the region was mainly controlled by the evolution of the Paleo-Asian Ocean, and it has been controlled by the integrated effect of the closure of the Mongol-Okhotsk Ocean and the subduction of the Paleo-Pacific Ocean Plate since the Mesozoic (Chen et al., 2011; Wu et al., 2011; Ouyang et al., 2013; Tang et al., 2014; Niu et al., 2017). Multiple tectonic systems make the Derbugan metallogenic belt a key tectonic unit of the composite orogenic system of the north-eastern Asian continental margin (Tang et al., 2018), and the coupling of tectonic-magma interactions in multiple stages has led to the formation of a dense area of Cu—Mo—Pb—Zn—Au—Ag polymetallic deposits in the Derbugan metallogenic belt. Chaganbulagen deposit is situated in

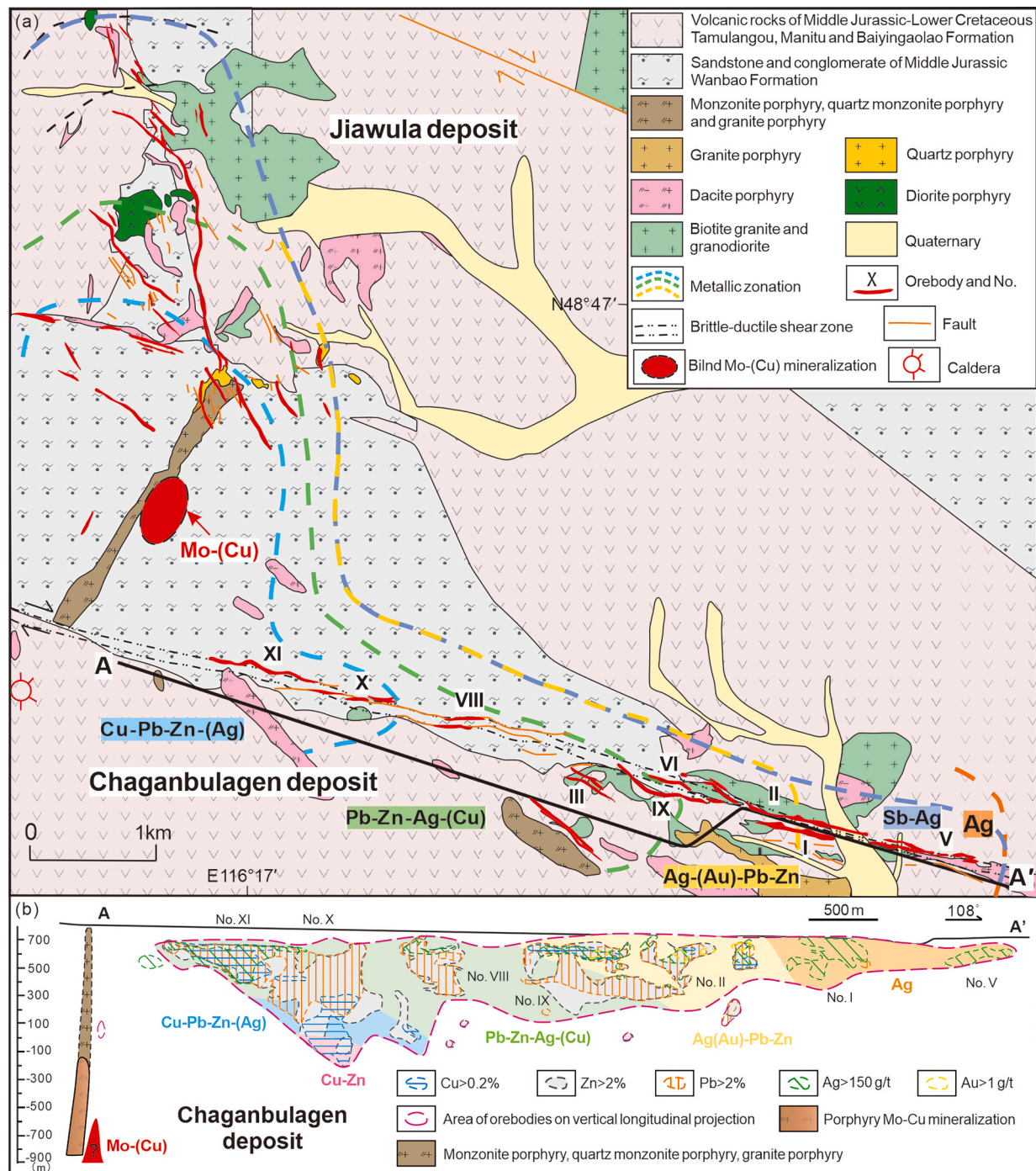


Fig. 2. (a) Location of Chaganbulagen-Jiawula ore field on the metallogenic belt. (b) Simplified vertical projection of porphyry Mo-(Cu) mineralization center and Chaganbulagen deposit and metal zonation. Distributional range of No. XI, X, VIII, IX, II, I and V orebodies in vertical projection of Chaganbulagen deposit. Anomalous metal enrichment areas were circled and partial metal cut-off grades were marked (modified from Hui et al., 2021).

the southwest section of the Derbugan metallogenic belt, which is a typical large Ag—Pb—Zn deposit on the belt (Fig. 1c).

Mesozoic volcanic sedimentary rocks and clastics rocks are widely distributed in the area, while Paleo-Neoproterozoic metasedimentary rocks and Paleozoic sedimentary rocks are only sporadically distributed. The major stratigraphy comprises Neoproterozoic slate, marble and metamorphic sandstone; as well as Mesozoic rocks including Middle Jurassic conglomerate and andesite, Upper Jurassic rhyolite, tuff, and trachyandesite, and Lower Cretaceous rhyolite, andesite and siltstone. Faults are developed in the metallogenic belt, represented by the NE-trending Derbugan fault. Secondary faults in the NW and NNE directions are also relatively developed. The NE- and NW-trending faults affect the volcanic magmatism and the spreading of a succession of metal deposits in the belt (Fig. 1c). The magmatic activities in the area are relatively strong, and the most closely related to metallogenesis are quartz porphyry, granite porphyry, diorite and granite of the Yanshanian (Cao and Liu, 2020). In addition, Neoproterozoic biotite granite, granodiorite, and quartz monzodiorite, and the Carboniferous-Permian hornblende granodiorite and monzonitic granite are distributed in the north-west side of the belt. A total of 29 medium- to large-scale deposits and 113 small-scale deposits and ore occurrences are reported in Derbugan metallogenic belt (Zhu et al., 2001). A number of Cu—Pb—Zn polymetallic deposits are located on the north-western side of the Derbugan fault. The Derbugan metallogenic belt can be divided into two periods of mineralization, mainly including: porphyry Cu—Mo deposits (e.g. Wunugetushan, Taipingchuan, Badaguan, Babayi) mineralized in the Early-Middle Jurassic, with an age of 205–177 Ma, and hydrothermal Pb—Zn—Ag(Mn) deposits (e.g. Erentaolegai, Jiawula, Chaganbulagen, Derbur), mainly in the Early Cretaceous, with an age of 141–120 Ma (Chen et al., 2010; She et al., 2010; Li et al., 2016; Zhang et al., 2016; Zhao et al., 2017).

3. Deposit geology

The exposed stratigraphy of the deposit is dominated by Middle Jurassic Wanbao Formation (J_2wb), Tamulangou Formation (J_2tm), Upper Jurassic Manitu Formation (J_3mn), and Lower Cretaceous Baiyingaolao Formation (K_1b) of Mesozoic era, and the lithologies are sandstone and conglomerate of Wanbao Formation, andesite of Tamulangou Formation, andesite of Manitu Formation, and rhyolite of Baiyingaolao Formation. Of these, the Wanbao and Tamulangou Formations are the mainly ore-hosting strata (Fig. 2). The deposit occurs as a radial fracture tectonic system with a quartz porphyry body as a convergence center (Li, 2016), and the Jiawula-Chaganbulagen brittle-ductile shear fracture zone presenting NW- and NWW-striking is the major ore-controlling tectonics, which is developed in the intermediate-basic volcanic rocks of Wanbao Formation and Tamulangou Formation, governing all the orebodies of the deposit to yield as vein-like. The intrusive rocks are primarily Hercynian granitoid (including granodiorite and biotite granite, dated at 258–252 Ma, Li et al., 2016; Niu et al., 2017; Hui et al., 2021), and Late Yanshanian porphyry groups intimately associated with the metallogenesis (dominated by quartz monzonite porphyry, monzonite porphyry and granite porphyry, dated at 139–146 Ma, Qin et al., 1996; Li, 2016; Yang et al., 2017).

Porphyry Mo (Cu) mineralization was detected in seven drill holes about 700–1500 m below surface between the Chaganbulagen large Ag—Pb—Zn deposit and Jiawula large Pb—Zn—Ag deposit (XYMCL, 2019; Hui et al., 2021). Chaganbulagen deposit has developed metallic mineralization zonation centered on deep porphyry Mo (Cu) mineralization, which transforms into Cu—Pb—Zn mineralization and Pb—Zn—Ag mineralization towards the distal and shallow parts of the deposit in sequence, and constituting a porphyry Mo (Cu) and epithermal Pb—Zn—Ag metallogenic system (Li, 2016; Yang et al., 2017; Hui et al., 2021). The principal orebodies of the deposit are No. I, II, III, V, VI, VIII, IX, X, XI. Among them, No. X, XI orebodies are the main Cu—Pb—Zn—(Ag) orebodies, which are closest to the porphyry Mo

(Cu) mineralization hydrothermal center (Fig. 2). Notably, No. X orebody shows strong Cu—Zn mineralization at the deep end and Pb—Zn—Ag mineralization at the shallow end. Ore minerals dominantly include chalcopyrite, arsenopyrite, pyrrhotite, sphalerite, pyrite, and galena. No. VIII, IX, and II are the main Pb—Zn—Ag—(Cu) orebodies, and are farther away from the porphyry Mo (Cu) mineralization. Ore minerals are mainly pyrite, sphalerite, and galena. No. I, and V are the main Ag orebodies, farthest away from the porphyry Mo (Cu) mineralization. Ag mineralization is usually interspersed as veinlets in the rhyolite. Ore minerals are sphalerite, pyrite, galena, and independent silver minerals are freibergite, pyrargyrite, argentite, stephanite, acanthite, and canfieldite. Silicification, carbonatization, sericitization, illitization, and chloritization are the common types of wall-rock alteration in the deposit, among which silicification is most relevant to mineralization (Li et al., 2016; Hui et al., 2021). Based on ore types, cross-cutting and replacement relationships of mineral assemblages, four metallogenic stages were classified: Cu—Zn, Cu—Pb—Zn, Pb—Zn—Ag, and Ag mineralization stages.

Cu—Zn mineralization stage is located 1.5–3 km from the porphyry Mo (Cu) mineralization hydrothermal center, and the mineralization range is concentrated in the deep part of No. X orebody (Fig. 2a). This stage consists mainly of dense massive chalcopyrite-sphalerite (SpI)-pyrrhotite-arsenopyrite sulfides with minor pyrite (Fig. 3a). Sphalerite (SpI) is generally coeval with pyrrhotite, and arsenopyrite shows cataclastic textures and replacement by sphalerite (SpI) (Fig. 3b). Meanwhile, sphalerite (SpI) contains abundant chalcopyrite inclusions (CcpI), and chalcopyrite (CcpI) is occasionally present as grains and fine veins (Fig. 3b and c).

Cu—Pb—Zn mineralization stage is 3–5.5 km away from the porphyry Mo (Cu) mineralization, and the mineralization range is represented by No. X, VIII, and II orebodies (Fig. 2b). This stage is characterized by the development of sphalerite (SpII)-pyrite-K-feldspar-quartz veins in rhyolite (Fig. 3d). Sphalerite in this stage includes both cluster-like (SpII-a) and fine vein-like (SpII-b) types. SpII-a contains numerous chalcopyrite inclusions (CcpII) of variable particle size and galena grains, and has replaced early euhedral pyrite, with pyrite skeletal crystals clearly visible (Fig. 3e). SpII-b cuts cataclastic pyrite as veins and coexists with galena (Fig. 3f).

Pb—Zn—Ag mineralization stage is 3–5.5 km away from the porphyry Mo (Cu) mineralization, and mineralization range mainly includes No. II orebody and the shallow part of No. X orebody (Fig. 2b). Large quantities of galena were precipitated during this stage, and primarily developing sphalerite (SpIII)-galena-quartz veins (Fig. 3g). SpIII-a mainly exhibits replacing small grains of euhedral arsenopyrite as veinlets, followed by symbiosis with pyrite and galena, with almost no chalcopyrite contained internally. While SpIII-b replaces early cassiterite, and galena and freibergite form intergrowth co-crystals endowed in SpIII-b, which also contains lots of chalcopyrite and freibergite inclusions (Fig. 3h). Chalcopyrite (CcpIII) corroded euhedral pyrite is locally observed and coexisted with SpIII-b and galena (Fig. 3i).

Ag mineralization stage is 5.5–8 km away from the porphyry Mo (Cu) mineralization, with a mineralization range of No. I, V orebodies (Fig. 2b). This stage exhibits veinlets of sphalerite (SpIV)-Ag-bearing minerals interspersed in the rhyolite (Fig. 3j). Sphalerite is coeval with galena and freibergite (Fig. 3k). Chalcopyrite (CcpIV) replaced by sphalerite (SpIV) and freibergite veinlets can be seen locally (Fig. 3l). Detailed paragenetic sequences of the principal minerals are presented in Fig. 4.

4. Sampling and analytical methods

Samples were collected from different mineralization zoning of Chaganbulagen deposit. A total of 21 representative samples were collected from four mineralization stages. Among them, samples from Cu—Zn stage were taken from drill hole ZK15007 at a depth of 836 m of No. X orebody, samples from Cu—Pb—Zn stage were taken from No. X,

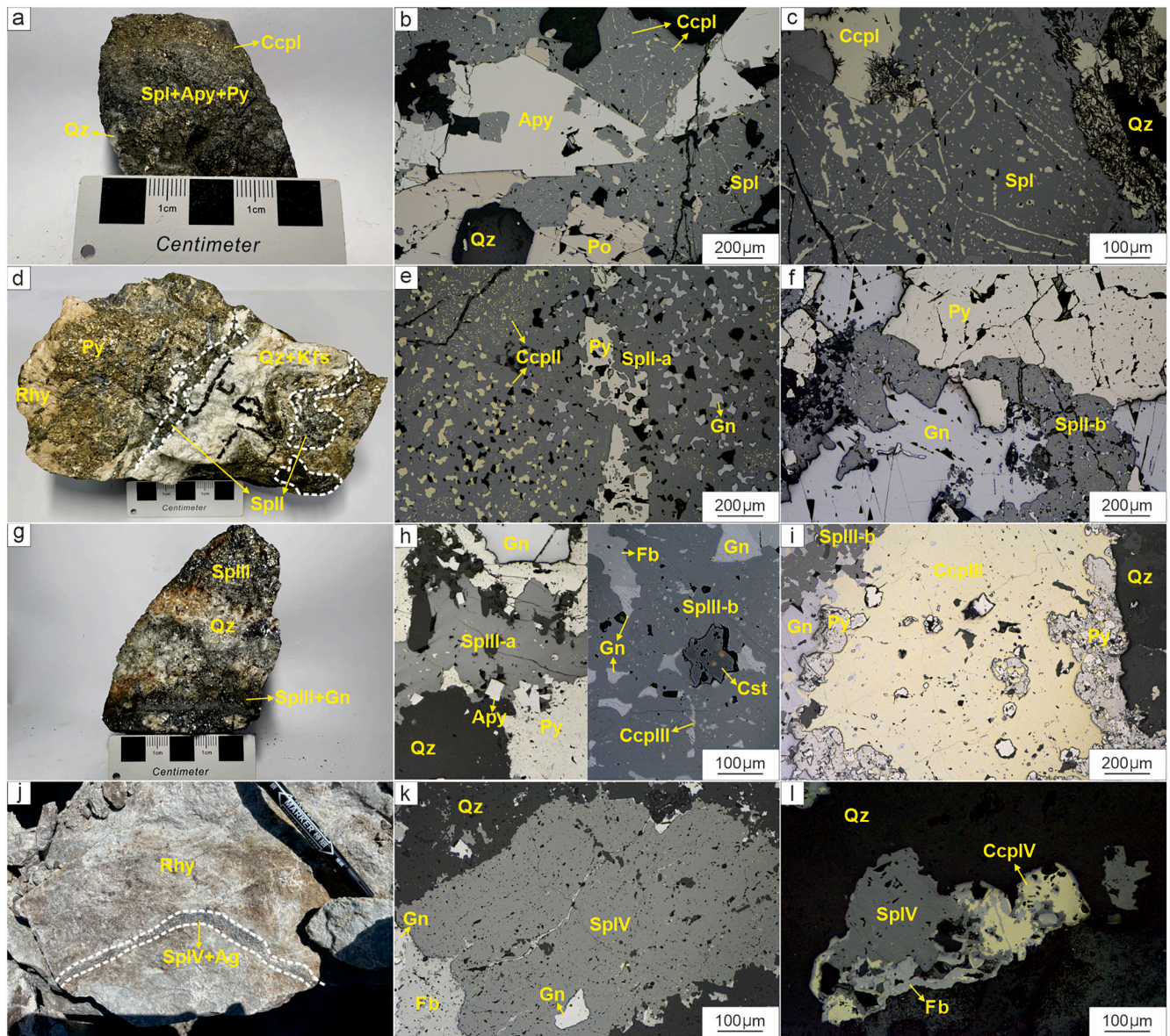


Fig. 3. Hand specimen photographs and microscopic photographs of representative mineral assemblages at different mineralization stages of Chaganbulagen deposit. (a) Massive chalcopyrite –sphalerite (Spl)-pyrite sulfide ores. (b) Sphalerite (Spl) is symbiotic with pyrrhotite and euhedral arsenopyrite, and arsenopyrite is corroded by sphalerite (Spl), Spl contains abundant chalcopyrite inclusions (Ccpl). (c) Numerous granular, fine-veined chalcopyrite (Ccpl) are contained in sphalerite (Spl). (d) Sphalerite (Spl)-pyrite-quartz-K-feldspar veins developed in rhyolite. (e) Sphalerite (SplII-a) contains abundant chalcopyrite inclusions (CcplII) of variable size and galena particles, and euhedral pyrite is replaced to form skeletal crystals. (f) Vein-like sphalerite (SplII-b) cross-cuts cataclastic pyrite and coexists with galena. (g) Sphalerite (SplIII)-galena-quartz veins. (h) SplIII-a replaces small grains of euhedral arsenopyrite and symbiosis with pyrite and galena; SplIII-b replaces early cassiterite, and galena and freibergite form intergrowth co-crystals endowed in SplIII-b, which also contains chalcopyrite (CcplIII) and freibergite inclusions. (i) Chalcopyrite (CcplIII) corroded euhedral pyrite is locally observed and coexisted with SplIII-b and galena. (j) Veinlets of Sphalerite (SplIV)-Ag-bearing minerals interspersed in the rhyolite. (k) Sphalerite is coeval with galena and freibergite. (l) Chalcopyrite (CcplIV) replaced by sphalerite (SplIV) and freibergite veinlets. Abbreviation: Apy-Arsenopyrite; Ccp-Chalcopyrite; Cst-Cassiterite; Fb-Freibergite; Gn-Galena; Kfs-K-feldspar; Po-Pyrrhotite; Py-Pyrite; Qz-Quartz; Rhy-Rhyolite; Sp-Sphalerite.

VIII, and II orebodies, samples from Pb-Zn-Ag stage were taken from No. II orebody and the shallow part of No. X orebody, samples from Ag stage were taken from No. I and V orebodies. Detailed in situ LA-ICP-MS analyses targeting minor and trace elements in sphalerite and chalcopyrite were carried out at Sample Solution Analytical Technology Co., Ltd., Wuhan, China. A total of 116 analytical points were selected from inclusion-free sulfide domains. All analytical procedures were conducted following standard operating protocols for sulfide mineral analysis. The specific instrument settings and analysis methods followed the procedures outlined in Zong et al. (2017). The laser ablation process employed a GeolasPro laser ablation system integrating a

COMPexPro102 ArF 193 nm excimer laser with MicroLas optical system, while elemental detection was accomplished using an Agilent 7900 ICP-MS. Throughout the analytical sequence, helium carrier gas and argon make-up gas were homogenized via a T-connector prior to enter the ICP. The laser ablation system is equipped with a signal smoothing device (Hu et al., 2015). The spot size and frequency of the laser beam for this analysis were 32 μm and 5 Hz, respectively. Trace element contents of sulfides were calibrated using the glass reference material NIST 610 for multiple external standards without internal standards (Liu et al., 2008), the sulfide reference material of MASS-1 (USGS) was used as a monitoring standard sample to verify the reliability of the calibration method.

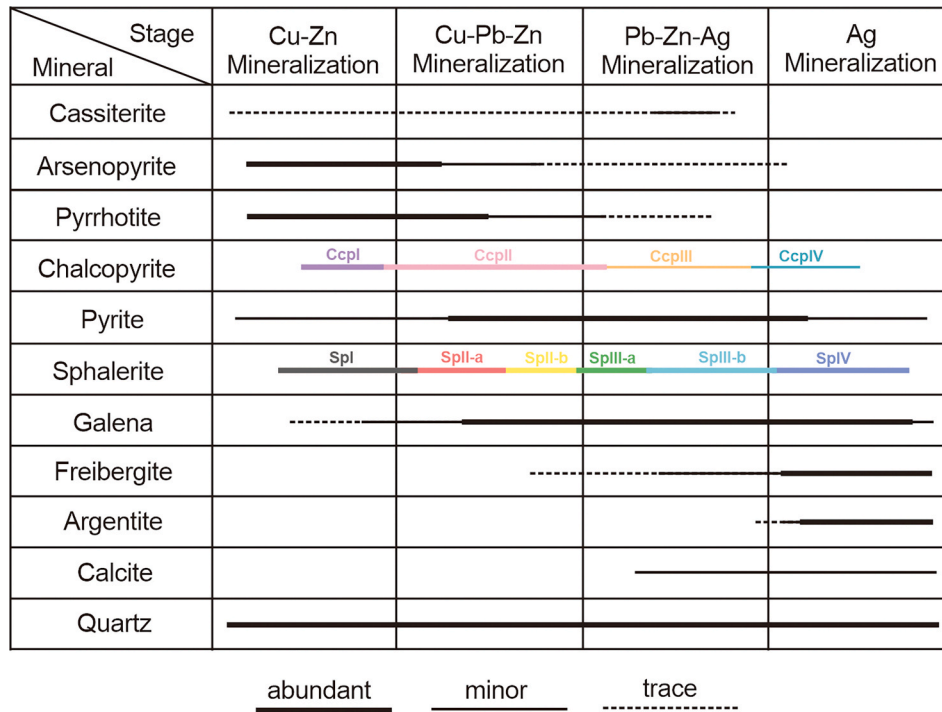


Fig. 4. Paragenetic sequences of the principal minerals in Chaganbulagen deposit.

Each analysis included 20–30 s of background acquisition followed by 50 s of sample data acquisition. Offline processing of the analyzed data was accomplished using the software ICPMSDataCal (Liu et al., 2008).

5. Results

5.1. Trace elements compositions of sphalerite

90 LA-ICP-MS spot analyses were completed from different generations of sphalerite and the analysis results are presented in Table 1 and Fig. 5. In general, sphalerite from Chaganbulagen deposit is featured by the enrichment of Fe, Zn, Mn, Cd, In, and deficit of Ga, Se, As, Sn. From the early Cu-Zn stage to the late Ag stage, Zn content of sphalerite is almost unchanged, and Se, Cd, Cu, Fe, Mn and In contents show a general decreasing trend, while Ag and Sb contents show an increasing trend.

SpI exhibits the highest Cd (av. 6763 ppm), In (av. 656 ppm), Se (av. 1401 ppm) contents, while its Cu content (av. 1125 ppm) is significantly higher than late stage SpIII-b (av. 220 ppm) and SpIV (580 ppm). Additionally, SpI contains elevated Fe (av. 68131 ppm), Mn (av. 2071 ppm), Pb (av. 141 ppm), minor Ag (av. 15.3 ppm), along with trace Ga (0.18–3.25 ppm), As (0–5.45 ppm), Sn (0.05–3.65 ppm), and Sb (0–0.67 ppm).

SpII-a and SpII-b show lower Fe contents (averaging 28,133 and 39387 ppm, respectively) compared to SpI. Their Ag, Mn, Sb, and Pb contents (averaging 62.0 and 61.5, 5226 and 2955, 4.22 and 1.75, 1108 and 455 ppm, respectively) are higher than SpI, with SpII-a displaying significantly elevated Mn and Pb relative to SpII-b. Conversely, Cd (6111 and 5509 ppm) and In (66.9 and 15.8 ppm) show marked decreases in both SpII-a and SpII-b.

SpIII-a demonstrates substantially lower Fe (av. 24051 ppm), Mn (av. 751 ppm), and Sn (av. 4.44 ppm) compared to SpIII-b (averaging 75821 ppm, 4218 ppm, 120 ppm, respectively), but higher Cu (3136 and 220 ppm), As (20.4 and 2.17 ppm), and Pb (2829 and 1.84 ppm). Both SpIII-a and SpIII-b exhibit steadily decreasing Cd (5192 and 4840 ppm) and In (10.4 and 3.52 ppm), with minor increases in Ag (80.4 and 105 ppm) and Sb (11.9 and 47.5 ppm).

SpIV contains the lowest Fe (av. 11104 ppm), Se (av. 10.9 ppm), Mn (av. 584 ppm), Cd (av. 4187 ppm), and In (av. 1.15 ppm) contents, but the highest concentrations of As (av. 270 ppm), Ag (av. 1963 ppm), Sb (av. 294 ppm), and Pb (av. 411 ppm) contents. Minor Se (av. 10.9 ppm), trace Ga (0.07–10.5 ppm) and Sn (0.07–4.26 ppm) are also present.

5.2. Trace elements compositions of chalcopyrite

A total of 26 in situ LA-ICP-MS spot analyses were carried out on chalcopyrite grains from different mineralization stages. The corresponding analysis results are shown in Table 2 and Fig. 5. Chalcopyrite in Chaganbulagen deposit is overall enriched in Zn, As, Ag, In, with deficits of Ga, Ge, Bi. In content shows a clear downward trend, and Ag and Sb show an upward trend.

Ccpl contains high levels of Zn (av. 13971 ppm), As (av. 325 ppm), Ag (av. 678 ppm), Cd (av. 143 ppm), In (av. 270 ppm), and Pb (av. 271 ppm). Notably, its In content (111–544 ppm) significantly surpasses that of CcpII (13.1–34.4 ppm), CcpIII (1.96–7.18 ppm), and CcpIV (0.09–0.98 ppm). Meanwhile, Ccpl contains minor Mn (av. 12.9 ppm), Sn (av. 31.3 ppm), Bi (av. 12.7 ppm), along with trace Ga (av. 0.39 ppm), Ge (av. 3.89 ppm), and Sb (av. 0.36 ppm).

CcpII shows reduced Zn (av. 13331 ppm), Ag (av. 339 ppm), Cd (av. 123 ppm), In (av. 23.6 ppm) and Pb (av. 12.7 ppm) compared to Ccpl, coupled with a sharp increase in Mn (av. 124 ppm). It maintains comparable As levels (av. 354 ppm) to Ccpl, with minor In (av. 23.6 ppm), Sn (av. 17.0 ppm), Pb (av. 12.7 ppm), and trace Ga (av. 1.71 ppm), Ge (av. 3.46 ppm), Sb (av. 4.71 ppm).

CcpIII demonstrates marked decreases in Zn (av. 7497 ppm), Mn (av. 17.5 ppm), Cd (av. 68.0 ppm), and In (av. 4.14 ppm) contents, but elevated As (av. 574 ppm), Ag (av. 627 ppm). And notably exceptional Sn content (429–1636 ppm, with an average of 926 ppm) is the highest of all chalcopyrite generations. Ga, Ge, Sb, Bi, and Pb contents remain broadly similar to CcpII levels.

CcpIV contains the maximum concentrations of Zn (av. 26319 ppm), Ag (av. 13157 ppm), Sb (av. 3286 ppm), and Pb (av. 470 ppm), paired with the lowest Cd (av. 92.0 ppm) and In (av. 0.36 ppm). Its As content (av. 401 ppm) slightly decreases compared to CcpIII, with trace Mn, Ga,

Table 1

The LA-ICP-MS analyses results of sphalerite from Chaganbulagen deposit.

Mineralization stage	Elements (ppm)	Fe	Cu	Zn	Ga	As	Se	Ag	Mn	Cd	In	Sn	Sb	Pb
Cu-Zn	Spl (n = 31)	76,349	574	570,097	3.25	0.72	531	7.08	2194	7742	506	1.66	0.10	1.39
		69,134	994	575,462	2.77	0.36	992	9.26	1978	7291	535	0.50	0.26	7.38
		70,470	817	573,286	2.05	bdl	2084	8.41	1910	6951	548	0.57	bdl	14.2
		73,694	322	572,324	2.36	0.27	507	5.23	2366	7202	554	0.41	0.03	0.94
		66,203	426	578,126	2.63	0.57	bdl	11.0	2479	7062	612	1.40	0.33	8.32
		80,827	591	562,693	1.77	5.45	1303	4.50	2342	7646	626	2.41	0.03	0.16
		78,122	4236	563,854	1.98	bdl	bdl	10.8	1921	7536	668	1.33	0.17	2.90
		79,820	4490	555,304	2.22	0.29	5464	12.9	2400	7177	514	3.65	0.10	1.81
		75,482	593	570,304	1.63	2.15	bdl	6.51	1714	6855	537	1.13	0.12	18.8
		62,598	724	569,269	2.64	bdl	12,987	26.5	2549	6591	721	0.18	0.16	26.1
		59,331	946	582,364	2.14	0.24	4227	36.9	2507	6407	744	0.06	0.16	32.6
		75,872	942	567,891	2.87	2.68	3167	27.6	1054	6302	819	0.12	0.31	23.0
		80,043	1113	564,299	2.61	0.92	1617	26.1	2104	5986	839	0.15	0.13	22.8
		68,693	1769	573,525	2.11	0.46	2964	32.3	2076	6222	932	0.16	0.11	33.0
		61,706	613	585,061	2.59	1.14	bdl	15.2	2542	6739	835	0.05	0.22	10.5
		61,480	2023	578,850	2.40	bdl	2454	35.5	2834	6258	1112	0.36	0.10	1603
		63,161	1417	580,387	1.58	0.06	bdl	31.0	2736	6075	987	0.14	0.67	1819
		66,288	680	576,278	2.02	0.03	4067	42.8	802	5740	1068	0.25	0.34	136
		58,824	647	587,548	3.24	0.30	bdl	38.3	2265	5984	1096	0.12	0.34	42.90
		25,835	675	622,302	0.18	bdl	bdl	21.6	159	7169	398	0.50	0.43	186
		28,008	728	614,579	0.49	bdl	bdl	24.8	165	6861	442	0.86	0.30	83.1
		70,892	789	574,466	1.02	0.24	bdl	4.30	2187	7339	545	0.68	0.04	0.83
		70,895	3883	571,883	2.77	0.50	247	7.89	1920	7096	526	1.84	0.12	290
		68,424	305	578,145	2.50	1.87	325	3.30	1373	6200	526	0.40	0.02	8.13
		72,854	318	573,175	1.56	bdl	195	2.07	2612	6983	547	0.46	0.05	0.10
		75,916	2729	568,131	1.68	0.53	bdl	11.3	2590	6810	548	0.19	bdl	0.10
		69,610	306	576,870	1.11	0.76	139	2.08	2569	6672	529	0.39	bdl	0.38
		72,551	339	572,708	0.33	0.86	bdl	2.57	2701	6730	500	0.40	0.03	0.70
		73,941	291	572,741	0.38	bdl	64.5	1.62	1977	6708	500	0.38	0.04	0.24
		73,779	290	573,296	1.21	0.21	49.8	2.49	2113	6501	504	0.48	0.08	0.65
		81,266	294	564,391	1.54	bdl	61.0	2.20	3051	6815	524	0.32	0.05	0.09
	Max	81,266	4490	622,302	3.25	5.45	12,987	42.8	3051	7742	1112	3.65	0.67	1819
	Min	25,835	290	555,304	0.18	bdl	bdl	1.62	159	5740	398	0.05	bdl	0.09
	Avg	68,131	1125	575,794	1.92	0.66	1401	15.3	2071	6763	656	0.70	0.16	141
Cu-Pb-Zn	SplII-a (n = 7)	53,675	3190	566,701	0.72	bdl	68.1	53.4	23,314	5570	61.6	4.00	9.25	233
		27,876	2467	616,412	1.68	bdl	bdl	38.7	4156	6801	71.6	0.24	0.17	201
		15,672	78.3	634,905	1.50	bdl	bdl	27.0	903	6209	66.6	1.04	5.82	76.7
		11,946	219	638,584	1.15	bdl	19.9	34.0	806	5937	76.8	1.17	4.44	2072
		33,017	13,547	599,862	2.42	0.60	bdl	130	2119	6258	66.9	0.27	1.90	4275
		26,021	6736	616,716	2.03	0.17	64.2	29.7	2690	5908	63.9	1.12	2.13	59.2
		28,725	9712	610,526	2.36	0.21	14.9	122	2590	6092	61.1	0.46	5.86	842
	Max	53,675	13,547	638,584	2.42	0.60	68.1	130	23,314	6801	76.8	4.00	9.25	4275
	Min	11,946	78.3	566,701	0.72	bdl	bdl	27.0	806	5570	61.1	0.24	0.17	59.2
	Avg	28,133	5135	611,958	1.69	0.14	23.9	62.0	5226	6111	66.9	1.18	4.22	1108
	SplII-b (n = 8)	65,616	1153	580,884	1.93	0.84	17.7	70.4	3827	5889	11.3	1.26	3.92	0.90
		66,927	394	578,208	2.15	bdl	bdl	27.8	4751	6323	14.0	0.14	0.10	1.92
		30,482	652	619,280	1.55	1.09	33.7	67.8	2378	5237	12.3	0.69	1.05	16.7
		31,043	15.1	617,866	3.09	0.60	bdl	10.4	2925	5583	14.7	0.15	0.32	2.98
		27,737	327	621,574	1.95	0.03	bdl	39.5	2339	5664	19.1	1.97	2.02	21.3
		36,902	4422	607,863	3.40	1.07	115.6	186	3100	5310	17.6	1.78	2.34	127
		31,880	16.1	618,180	1.45	0.07	bdl	5.43	2736	4687	18.1	0.20	0.87	3.81
	Max	66,927	4422	621,574	3.40	1.09	116	186	4751	6323	19.3	1.97	3.92	3467
	Min	24,506	15.1	578,208	1.45	bdl	bdl	5.43	1585	4687	11.3	0.14	0.10	0.90
	Avg	39,387	1222	608,125	2.20	0.56	27.9	61.5	2955	5509	15.8	0.84	1.75	455
Pb-Zn-Ag	SplIII-a (n = 10)	34,321	2566	615,699	2.89	11.7	bdl	55.9	1014	5811	5.96	1.32	2.74	6.53
		21,689	227	630,698	2.20	18.2	37.4	58.2	910	4667	4.67	4.31	10.5	273
		24,465	3784	616,437	2.56	22.5	bdl	122	793	4859	6.51	7.31	18.2	10,925
		46,232	6919	597,935	5.50	22.8	0.79	80.2	1720	4646	11.0	2.67	0.66	8.63
		18,662	1192	633,572	0.55	15.4	bdl	49.0	524	5947	12.7	9.50	11.5	51.0
		16,345	509	635,296	0.92	33.9	18.4	96.0	431	5430	15.6	4.98	20.8	417
		16,989	391	633,616	3.14	19.4	2.61	49.0	487	5317	12.3	2.08	12.3	1527
		16,036	771	624,678	2.26	17.5	9.21	64.8	435	5673	12.3	3.71	17.5	14,986
		11,900	75.4	642,314	0.74	16.6	17.0	85.3	324	4289	10.8	6.51	25.0	81.7
		33,874	14,930	602,771	0.54	25.7	26.1	144	869	5281	12.2	1.99	0.23	13.8
	Max	46,232	14,930	642,314	5.50	33.9	37.4	144	1720	5947	15.6	9.50	25.0	14,986
	Min	11,900	75.4	597,935	0.54	11.7	bdl	49.0	324	4289	4.67	1.32	0.23	6.53
	Avg	24,051	3136	623,301	2.13	20.4	11.1	80.4	751	5192	10.4	4.44	11.9	2829
	SplIII-b (n = 20)	84,296	302	561,985	12.68	0.78	bdl	8.38	3876	5511	9.14	1.84	0.04	0.01
		83,753	120	561,638	8.60	0.14	bdl	6.76	3963	5441	0.69	70.3	0.09	0.28
		83,231	123	562,516	12.84	0.39	37.8	8.50	3990	5229	1.10	79.5	0.15	1.01
		81,242	189	564,683	9.84	0.32	90.3	10.3	4488	5258	1.45	158	0.18	0.92
		76,838	22.2	569,735	3.90	bdl	150	2.30	3873	5221	2.63	3.09	0.11	0.10
		76,209	140	569,521	13.18	0.60	bdl	11.2	4014	4990	1.86	105	0.03	0.26

(continued on next page)

Table 1 (continued)

Mineralization stage	Elements (ppm)	Fe	Cu	Zn	Ga	As	Se	Ag	Mn	Cd	In	Sn	Sb	Pb
Ag	Max Min Avg SpIV (n = 14)	74,774	63.3	572,210	5.31	0.80	bdl	6.32	3870	5144	4.35	10.1	0.04	0.35
		73,446	50.5	574,861	5.45	bdl	63.4	9.53	4754	4261	2.51	28.5	0.62	1.10
		75,121	238	571,910	3.09	0.08	bdl	41.4	5010	4342	9.99	6.57	0.62	0.98
		70,218	14.7	576,561	8.06	0.73	47.0	3.82	4822	3909	5.49	1.21	0.83	0.29
		79,137	49.7	567,776	5.24	0.40	bdl	40.6	5239	5132	13.6	30.3	0.43	0.57
		72,147	206	574,493	6.18	0.55	17.5	61.9	5010	4155	2.29	189	1.41	1.42
		78,659	10.5	566,992	7.54	0.57	bdl	4.23	5221	4826	0.23	0.62	2.57	1.20
		77,623	28.6	568,187	11.7	0.03	bdl	21.9	5015	5110	0.16	1.95	10.6	4.52
		78,936	61.7	566,083	13.9	0.45	75.8	199	5452	5176	0.52	80.5	2.67	11.3
		79,979	2099	559,274	13.0	10.6	86.3	1522	5388	5311	7.12	1602	855	4.80
		67,664	8.85	581,625	7.43	0.22	82.4	4.18	1737	4573	0.19	1.24	0.04	0.26
		69,043	7.76	580,145	5.76	bdl	bdl	3.69	2832	4341	0.39	0.55	0.69	0.82
		55,119	52.4	595,939	6.81	0.85	bdl	75.0	1195	4047	6.74	21.2	70.8	3.31
		78,985	608	567,164	9.90	25.8	14.8	52.4	4612	4827	0.05	2.22	2.92	3.20
		84,296	2099	595,939	13.9	25.8	150	1522	5452	5511	13.6	1602	855	11.3
		55,119	7.76	559,274	3.09	bdl	bdl	2.30	1195	3909	0.05	0.55	0.03	0.01
		75,821	220	570,665	8.52	2.17	33.3	105	4218	4840	3.52	120	47.5	1.84
		3603	17.0	650,086	0.53	0.36	17.6	37.9	17.8	7071	0.14	0.11	21.7	4.66
		13,773	123	638,143	0.59	1.29	bdl	51.7	1356	5788	0.56	0.26	32.8	21.9
		3005	32.7	649,293	1.52	3.30	24.8	61.8	6.66	7408	0.36	2.87	44.2	7.43
		7393	30.7	646,655	0.07	3.47	bdl	44.7	294	4363	0.16	0.11	41.5	31.9
		9385	599	641,852	0.13	1.96	44.4	107	825	5555	0.13	0.72	62.9	261
		8962	89.4	646,204	0.30	0.73	23.9	26.4	95.6	4145	0.31	0.50	15.0	7.94
		6087	65.6	648,257	3.09	4.37	bdl	97.1	34.7	3684	2.44	1.89	80.6	1230
		9696	56.6	644,793	1.02	2.06	18.5	95.0	14.2	3990	0.49	1.12	77.1	7.06
		3856	6749	614,446	1.26	3749	bdl	26,876	121	302	0.04	0.89	3656	4067
		13,555	2.42	642,266	0.40	0.77	bdl	3.09	487	2500	0.33	0.07	2.81	0.47
		3990	41.1	653,027	1.06	5.71	bdl	57.6	6.23	2576	0.67	3.15	62.9	8.99
		9427	9.66	646,588	0.39	1.34	bdl	6.16	313	2500	1.03	1.36	12.0	111
		27,519	146	623,785	4.29	bdl	bdl	8.36	2252	4223	5.21	3.41	2.52	0.63
		35,210	160	615,249	10.5	0.43	23.4	9.40	2348	4518	4.18	4.26	0.11	0.70
		35,210	6749	653,027	10.5	3749	44.4	26,876	2348	7408	5.21	4.26	3656	4067
		3005	2.42	614,446	0.07	bdl	bdl	3.09	6.23	302	0.04	0.07	0.11	0.47
		11,104	580	640,046	1.80	270	10.9	1963	584	4187	1.15	1.48	294	411

Abbreviations: Sp =Sphalerite. Notes: bdl, below detection limit; Non-detectable samples were assigned zero values in statistical computations.

Ge, Bi, and Sn (2.16–15.3 ppm) detected.

5.3. The occurrence of indium and cadmium

Based on the above results, indium and cadmium are relatively enriched in Chaganbulagen deposit, with their specific occurrences as follows (Fig. 6). Indium is predominantly endowed in sphalerite (0.04–1112 ppm, averaging 235 ppm), followed by chalcopyrite (0.09–544 ppm, averaging 70.1 ppm), and virtually absent in pyrite (0–1.07 ppm, averaging 0.1 ppm) (authors' unpublished data). Additionally, independent minerals phases of indium as well as micro- and nano-sized mineral encapsulations of indium have not been detected in Chaganbulagen deposit. Similar to indium, cadmium also occurs primarily in sphalerite (302–7742 ppm, averaging 5598 ppm), with a secondary concentration in chalcopyrite (2.16–740 ppm, averaging 105 ppm) and almost none in pyrite (0–13.5 ppm, averaging 1.46 ppm).

6. Discussion

6.1. Correlation trend and element substitution

6.1.1. Correlation trend and element substitution in sphalerite

Sphalerite, as a significant host mineral for multiple rare and dispersed trace elements, has been widely discussed in terms of the distribution characteristics and substitution mechanisms of its trace elements (Xu et al., 2021b; Torró et al., 2021; Liu et al., 2023). Numerous studies have shown that divalent elements with similar ionic radius (Fe, Mn, Cd, etc.) generally replace Zn in the sphalerite lattice directly by isomorphism, while trivalent and quaternary elements (In, Ga, Ge, Sn, etc.) are usually substituted with monovalent elements (Ag, Cu, etc.) for Zn through elemental coupling substitution mechanisms (e.g., $2\text{Zn}^{2+} \leftrightarrow \text{Cu}^{+} + \text{In}^{3+}$, $3\text{Zn}^{2+} \leftrightarrow \text{Ag}^{+} + \text{Cu}^{+} + \text{Sn}^{4+}$) (Cook et al., 2009; Li

et al., 2020; Liu et al., 2022). In the LA-ICP-MS laser ablation signal map, the time signal intensity spectrum shows a smooth straight line, indicating that the trace elements are mainly embedded in the minerals by isomorphism; when the spectrum appears multiple wave peaks, showing a curve with a large fluctuation amplitude, suggesting that the trace elements mainly exist as micro-inclusions (Cook et al., 2009; Belissont et al., 2014).

In Chaganbulagen deposit, the signal profiles of sphalerite (Fig. 7) indicates that Fe, Mn, Cd, In and Cu elements exhibit flat straight lines or curves parallel to Zn, implying that the Fe, Mn, Cd, In and Cu enter the sphalerite lattice mainly by isomorphisms (Cook et al., 2009; Ye et al., 2012; Leng and Qi, 2017). During Cu–Zn stage, the small fluctuation of Cu element with the increase of stripping time (Fig. 7a) indicates that Cu not only enters into sphalerite by isomorphism, but also exists in sphalerite as micro-inclusions in the early stage, which is consistent with the occurrence of a large number of chalcopyrite inclusions inside the early stage Sphalerite (SpI) in the results of microscopic observation (Fig. 3b and c). In Cu–Pb–Zn stage, influenced by the chalcopyrite inclusions in SpII (Fig. 3e), the signals of Cu element continue to show small fluctuations, and the anomalous fluctuations of Ag and Pb elements show the presence of mineral micro-inclusions of Ag and Pb (Fig. 7b). Upon entering the Pb–Zn–Ag stage, the signal of Sn becomes intense, presenting a flat straight line like most of the elements; Ag shows a straight line generally but with anomalous fluctuations locally, and Pb shows a wave peak (Fig. 7c), implying that Sn enters sphalerite through isomorphism, Ag is dominated by isomorphism and also influenced by micro-inclusions, while Pb exists as micro-inclusions. At the latest Ag stage, Cu, Ag, Sb, and Pb exhibit unstable signals with synchronous variations (Fig. 7d), indicating that they jointly formed freibergite micro-inclusions and were hosted in the sphalerite at this time. This is further supported by the petrographic observation that the late Ag stage of chaganbulagen deposit hosts abundant freibergite (Fig. 3k

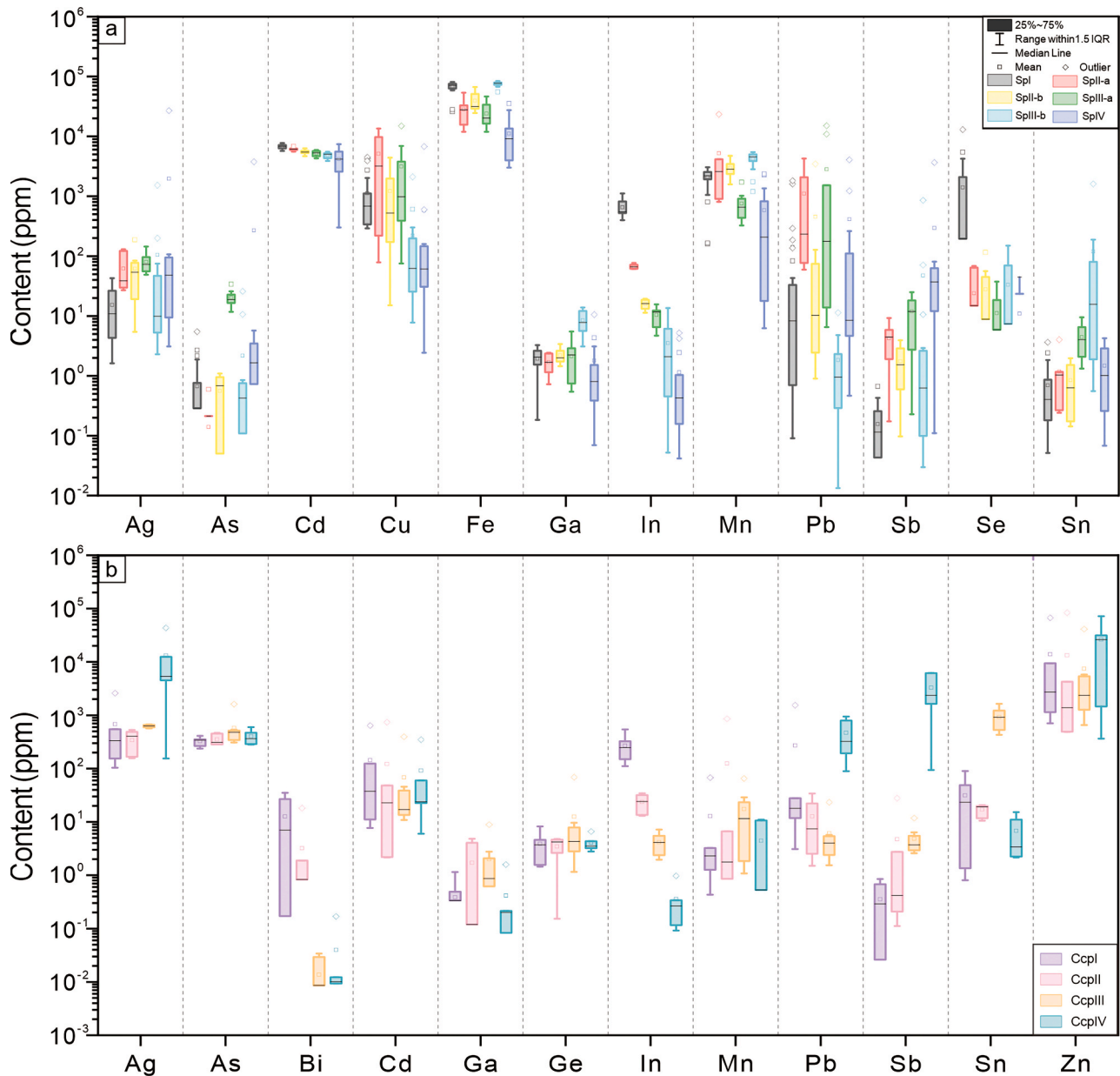


Fig. 5. Box-whisker plots of main trace elements in different generations of sphalerite and chalcopyrite in Chaganbulagen deposit. Abbreviation: Sp- sphalerite; Ccp- chalcopyrite.

and I). In the trace element correlation diagrams (Fig. 8), Fe, Mn, and Cd of sphalerite shows obvious negative correlation with Zn (Fig. 8a), which confirms the substitution of $\text{Zn}^{2+} \leftrightarrow \text{Fe}^{2+}$, Mn^{2+} , Cd^{2+} . The Cu-Ag and Cu-In correlations reveal the fluid evolution, illustrating a trend of decreasing Cu and In content alongside increasing Ag content from early to late stages (Fig. 8b and c). Meanwhile, sphalerite (SpI) of the early Cu-Zn stage exhibits a positive correlation between Cu and In (Fig. 8c), implying that In in the deposit mainly combines with Cu^+ to replace Zn in sphalerite by a substitution mechanism of $\text{Cu}^+ + \text{In}^{3+} \leftrightarrow 2\text{Zn}^{2+}$; however, it weakens in the late Pb-Zn-Ag and Ag stages, implying the substitution mechanism mainly functions in the early stage. Sn-In (Fig. 8d) and Cu-Sn (Fig. 8e) data points are relatively scattered, but it can be observed that most of the SpIII and SpIV data points are distributed near the $\text{Sn}/\text{In} = 1$ line, and the majority of SpIII-b data points fall on the $\text{Cu}/\text{Sn} = 2$ line, suggesting that $\text{In}^{3+} + \text{Cu}^+ + \text{Sn}^{4+} + \square$ (vacancy) $\leftrightarrow 4\text{Zn}^{2+}$ (Frenzel et al., 2014), $\text{In}^{3+} + \text{Sn}^{3+} + \square$ (vacancy) $\leftrightarrow 3\text{Zn}^{2+}$ (Belissant et al., 2014) and $2\text{Cu}^+ + \text{Sn}^{4+} \leftrightarrow 3\text{Zn}^{2+}$ substitution may play a certain role at Pb-Zn-Ag and Ag stages, at least for sphalerite

(SpIII-b) with relatively high Sn content (up to 1602 ppm) (Xu et al., 2021b). In the Sn-In correlation (Fig. 8d), the data points of SpIII-b beyond the $\text{Sn}/\text{In} = 1$ line are probably affected by cassiterite. In the late Ag stage, the participation of Ag^+ enhances the possibility of complementary coupled substitutions involving In, Cu, Ag, Sn, and Zn, such as $\text{In}^{3+} + (\text{Cu}, \text{Ag})^+ + \text{Sn}^{2+} \leftrightarrow 3\text{Zn}^{2+}$ or $\text{In}^{3+} + (\text{Cu}, \text{Ag})^+ + \text{Sn}^{4+} \leftrightarrow 4\text{Zn}^{2+}$ (Frenzel et al., 2014; Xu et al., 2021b). However, since the occurrence of multiple oxidation states of Sn (Sn^{2+} , Sn^{3+} , and Sn^{4+}) in sphalerite is currently ambiguous, the substitution of Sn element requires further research. Several of the data plotted on the outside of the positive correlation ratio line may be influenced by the chalcopyrite and freibergite inclusions within the sphalerite (Fig. 8b, c and f).

6.1.2. Correlation trend and element substitution in chalcopyrite

Compared to sphalerite, chalcopyrite is less conducive to the entry of trace elements (Cook et al., 2009; George et al., 2016; George et al., 2018). However, chalcopyrite in Chaganbulagen deposit is significantly enriched in Zn, As, Ag, In, and Sn, and deficient in Ga, Ge, and Bi (Fig. 5).

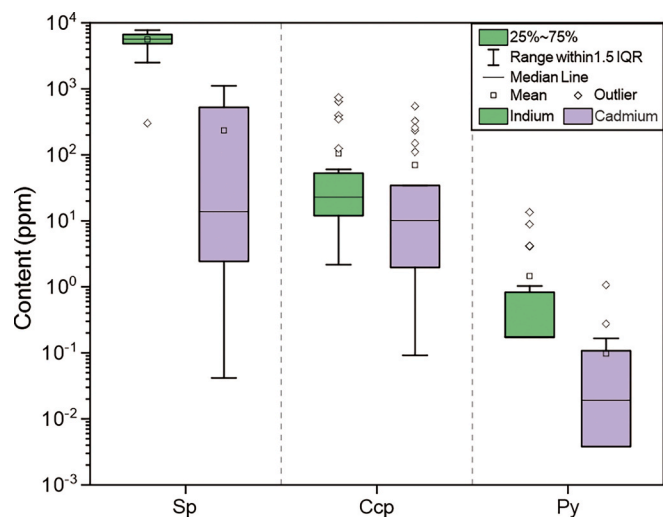


Fig. 6. Box-whisker plots of indium and cadmium contents in sphalerite, chalcopyrite, and pyrite of Chaganbulagen deposit. Abbreviation: Sp- sphalerite; Ccp-chalcopyrite; Py- pyrite.

LA-ICP-MS analysis shows that the majority of Zn content in chalcopyrite is above 2000 ppm, which is mainly endowed in chalcopyrite as sphalerite micro-inclusions (Fig. 7g and h), and only displaces Fe^{2+}

under high-temperature conditions in Cu–Pb–Zn stage by isomorphism ($\text{Zn}^{2+} \leftrightarrow \text{Fe}^{2+}$) (Fig. 7e and f). The average Ag content reaches 2971 ppm, and the stripping signals are relatively stable in chalcopyrite during Cu–Zn, Cu–Pb–Zn, and Pb–Zn–Ag stages (Fig. 7e–g), suggesting that Ag enters into the chalcopyrite lattice mainly through isomorphism. In the late Ag stage, Ag displays synchronous abnormal signals with Sb and Pb indicating the presence of micro-inclusions of freibergite or argentite (Fig. 7h). This implies that Ag^+ preferentially enters the chalcopyrite lattice under high-temperature conditions, and Ag is more likely to form micro-inclusions in low-temperature hydrothermal environments. From the signal profiles of the Cu–Zn and Cu–Pb–Zn stages, it can be obtained that In element enters into chalcopyrite by isomorphism (Fig. 7e and f). In the binary correlation diagram (Fig. 8), Cd–Zn positive correlation ($R^2 = 0.63$) indicates a simple substitution between Cd and Zn (Fig. 8g). From Cu–Zn stage to Ag stage, the In content decreases and Ag content increases in chalcopyrite (Table 2), and In and Ag show a negative correlation ($R^2 = 0.43$) (Fig. 8h). Since chalcopyrite and sphalerite have similar crystal structures, it is speculated that there may also be a coupling substitution mechanism between In and Ag in chalcopyrite. Sn element is obviously enriched in CcpIII of Pb–Zn–Ag stage (429–1636 ppm, averaging 926 ppm), and the signal profile of Sn becomes stronger and displays a smooth straight line (Fig. 7g) like that of SpIII, and meanwhile, In and Sn exhibit a strong positive correlation in CcpIII ($R^2 = 0.99$) (Fig. 8i), which further demonstrates that Sn enters into the chalcopyrite lattice through isomorphism and that a substitution relationship exists between In and

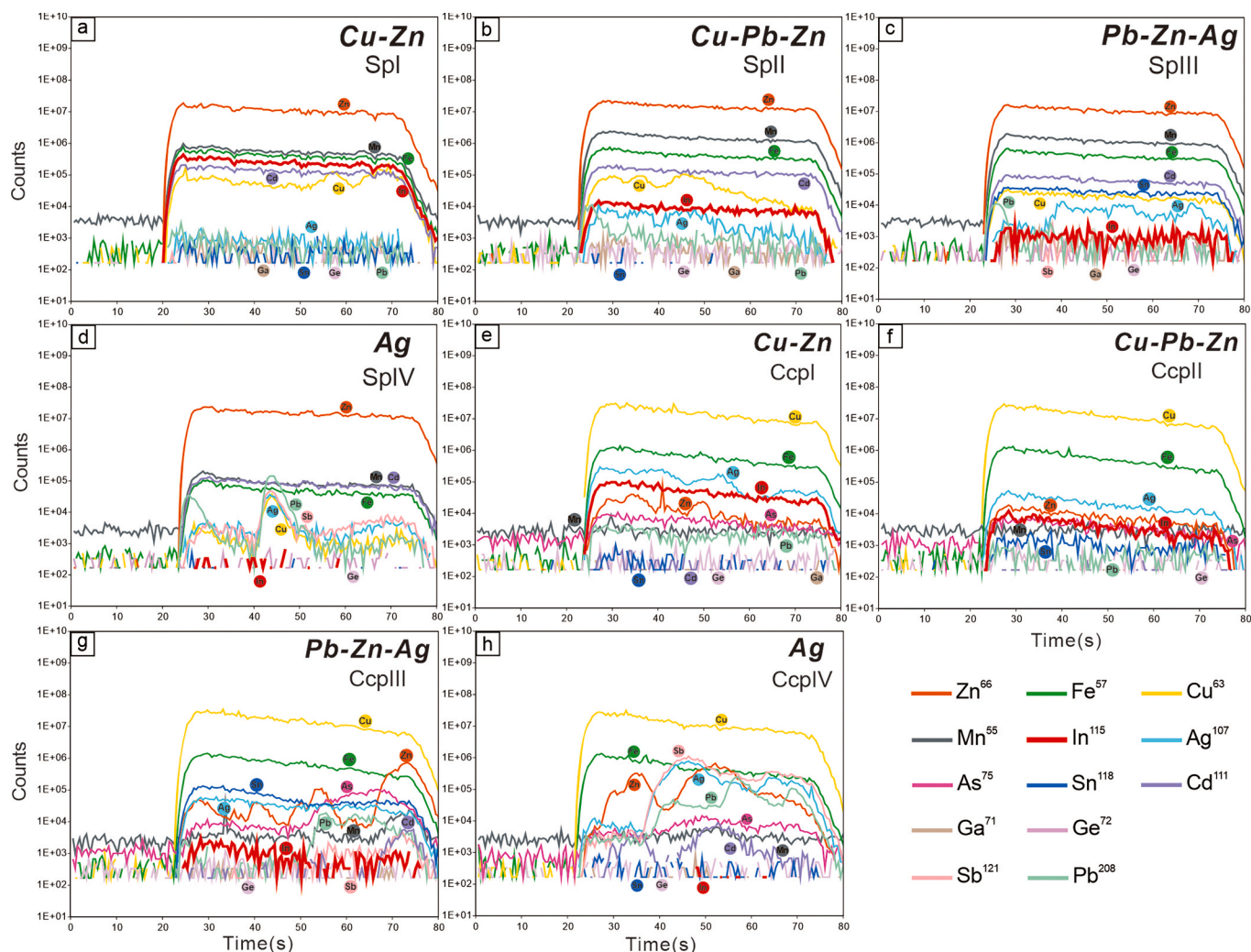


Fig. 7. Representative LA-ICP-MS time-resolved depth profiles for selected elements in sphalerite and chalcopyrite.

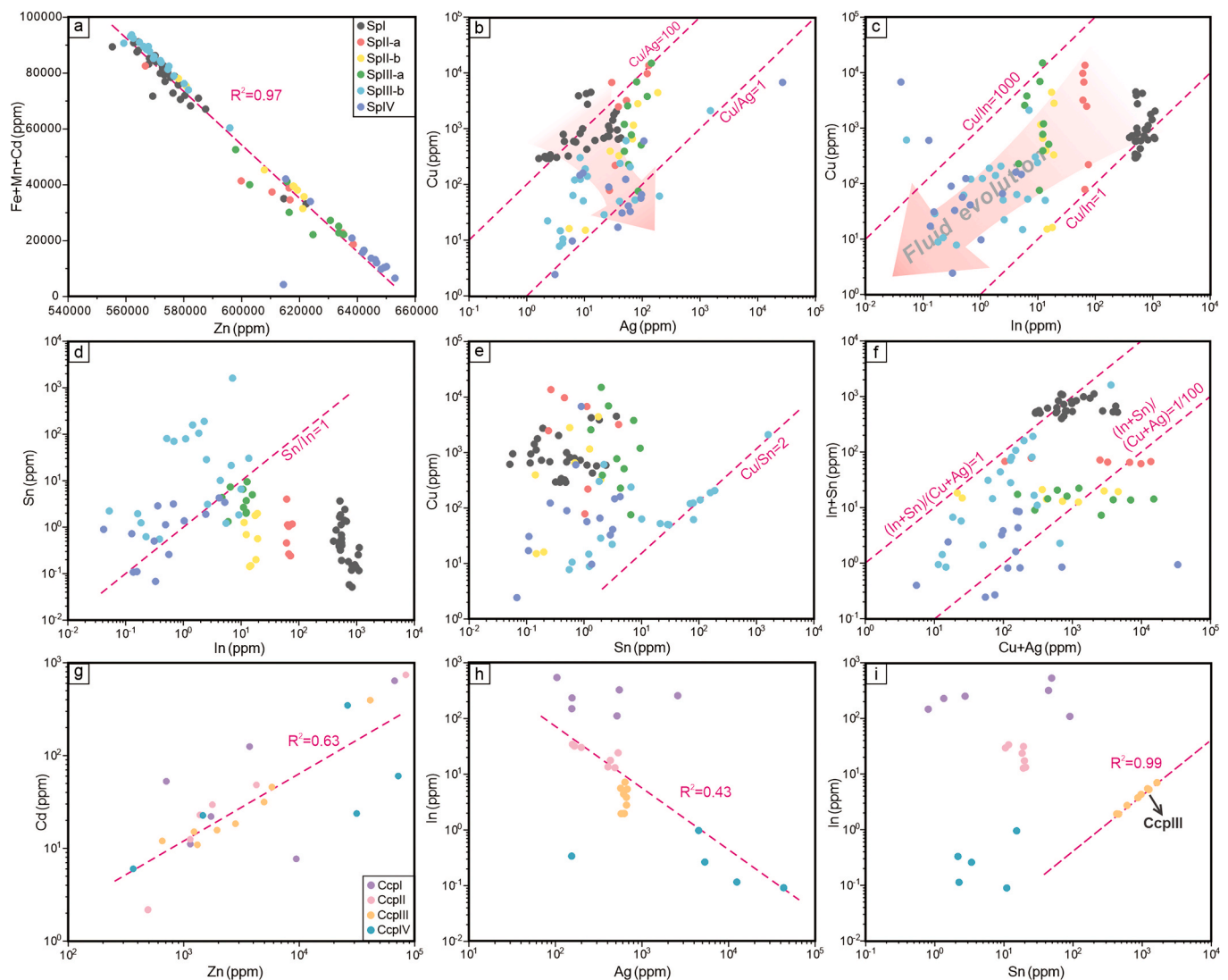


Fig. 8. Correlation plots of the major and trace element in sphalerite (a-f) and chalcopyrite (g-i): (a) (Fe + Mn + Cd) vs. Zn; (b) Cu vs. Ag; (c) Cu vs. In; (d) Sn vs. In; (e) Cu vs. Sn; (f) (In + Sn) vs. (Cu + Ag). (g) Cd vs. Zn; (h) In vs. Ag; (i) In vs. Sn.

Sn. Furthermore, the petrographic facts of the discovery of early residual fine-grained cassiterite in this stage (Fig. 3h) provide evidence for these conclusions as well. Additionally, the In-Sn correlation is not significant in chalcopyrite in all stages except for CcpIII of Pb-Zn-Ag stage, which is due to the fact that the other stages are not enriched in Sn. It can be inferred that when the fluid is Sn-enriched, the substitution relationship between In and Sn exists and may be the primary substitution mechanism for In entering the chalcopyrite and sphalerite. Overall, the occurrence of trace element in chalcopyrite is similarly controlled by temperature evolution, as well as by the influence of different ions concentrations in multi-stage solutions.

6.2. Indication of metallogenic temperature by trace element composition in sphalerite

The changes in various trace elements in sphalerite can reflect changes in the physical and chemical conditions of mineralization (Han, 1994; Ye et al., 2012; Deditius et al., 2014; Belissont et al., 2016). Previous studies have revealed that the elemental compositions of Fe, Mn, Co, Cd, In, Ga and Ge can effectively indicate the metallogenic temperature (Möller, 1987; Kelley et al., 2004; Cook et al., 2009; Ye et al., 2012; Frenzel et al., 2016; Zhang et al., 2022).

There are obvious differences in the color of sphalerite at different

metallogenic temperatures, forming darker sphalerite relatively enriched in Fe, Mn, and In at high temperatures with lower Ga/In ratios, and lighter sphalerite enriched in Ga and Ge at low temperatures with higher Ga/In ratios (Liu et al., 1984; Ye et al., 2012). In Chaganbulagen deposit, Fe content of sphalerite is higher in Cu-Zn stage (averaging 68131 ppm) compared to Cu-Pb-Zn (averaging 34135 ppm), Pb-Zn-Ag (averaging 58564 ppm), and Ag stage (averaging 11104 ppm), and the Ga/In ratio of sphalerite varies from early Cu-Zn and Cu-Pb-Zn stages (0–0.21, average 0.03) to increase in late Pb-Zn-Ag and Ag stages (0.04–189.75, average 10.98), suggesting a gradual decrease in temperature with the metallogenesis. The Zn/Cd ratios of sphalerite from Chaganbulagen deposit are largely between 100 and 500, suggesting that sphalerite crystallizes mainly under intermediate-temperature conditions of 200–300 °C (Han, et al., 1994; Ye et al., 2023). Notably, Chaganbulagen deposit exhibits moderate selenium (Se) enrichment, a feature also documented in other porphyry and epithermal systems (e.g., the Shuangjianzishan deposit, China, Zhai et al., 2020). The sphalerite in Cu-Zn stage contains Se-bearing inclusions, while show a general decreasing trend from Cu-Zn stage to Ag stage, this trend may further reflect cooling of the ore-forming fluids. The sphalerite geothermometer (GGIMFis) proposed by Frenzel et al. (2016) based on the content of some trace elements in sphalerite was used to estimate the metallogenic temperature of the deposit, and the metallogenic temperatures of

Table 2

The LA-ICP-MS analyses results of chalcopyrite from Chaganbulagen deposit.

Mineralization stage	Elements (ppm)	Zn	Mn	Ga	Ge	As	Ag	Cd	In	Sn	Sb	Bi	Pb
Cu-Zn	Ccpl (n = 6)	9437	0.43	bdl	3.03	354	104	7.68	544	49.0	0.68	2.13	11.8
		3719	2.54	bdl	4.61	266	515	125	111	89.9	0.40	bdl	3.10
		707	1.27	0.49	1.45	238	548	52.7	326	44.1	0.85	0.17	1545
		67,097	67.7	0.42	1.56	331	156	639	234	1.35	0.03	26.8	27.8
		1142	3.25	0.26	8.25	411	155	11.1	150	0.80	0.18	35.0	22.0
		1723	2.06	1.15	4.41	351	2590	22.0	257	2.75	bdl	11.9	13.9
	Max	67,097	67.7	1.15	8.25	411	2590	639	544	89.9	0.85	35.0	1545
	Min	707	0.43	bdl	1.45	238	104	7.68	111	0.80	bdl	bdl	3.10
	Avg	13,971	12.9	0.39	3.89	325	678	143	270	31.3	0.36	12.7	271
	Cu-Pb-Zn	CcplII (n = 7)	4259	6.68	0.12	4.35	285	431	48.2	17.7	19.8	0.21	1.89
1380			0.87	bdl	4.81	284	405	22.9	13.5	20.6	0.11	0.83	1.50
1137			bdl	bdl	3.46	290	530	12.5	24.2	18.4	0.42	1.66	2.53
1770			2.39	bdl	4.20	310	487	29.5	13.1	19.1	2.75	18.1	22.2
492			1.78	2.98	2.62	453	199	2.19	30.2	10.5	0.34	bdl	7.43
83,793			857	4.05	4.65	387	157	740	34.4	11.7	1.39	bdl	18.2
Max		489	0.85	4.84	0.15	466	166	2.16	32.0	19.2	27.7	bdl	34.2
Min		83,793	857	4.84	4.81	466	530	740	34.4	20.6	27.7	18.1	34.2
Avg		489	bdl	bdl	0.15	284	157	2.16	13.1	10.5	0.11	bdl	1.50
Max		13,331	124	1.71	3.46	354	339	123	23.6	17.0	4.71	3.21	12.7
Pb-Zn-Ag	CcplIII (n = 8)	5809	7.94	1.37	2.84	339	636	45.7	7.18	1636	2.90	0.03	1.54
		2807	15.1	2.78	2.76	1618	660	18.4	3.83	865	6.36	0.03	23.4
		4967	17.9	0.93	9.63	346	662	31.5	2.79	606	4.65	bdl	5.70
		652	1.29	0.62	1.16	309	624	12.0	1.97	429	2.96	bdl	4.67
		1229	28.8	0.81	3.67	495	612	15.1	4.42	966	3.94	0.02	2.51
		41,253	65.2	0.62	6.16	479	676	395	5.39	1253	2.59	bdl	2.27
	Max	1947	1.08	bdl	4.98	533	568	15.7	5.58	1197	11.8	0.03	5.14
	Min	1313	2.40	8.90	69.4	475	575	10.9	1.96	456	3.46	bdl	3.40
	Avg	41,253	65.2	8.90	69.4	1618	676	395	7.18	1636	11.8	0.03	23.4
	Max	652	1.08	bdl	1.16	309	568	10.9	1.96	429	2.59	bdl	1.54
Min	7497	17.5	2.00	12.6	574	627	68.0	4.14	926	4.84	0.01	6.08	
Ag	CcplIV (n = 5)	1465	bdl	bdl	4.35	365	4513	22.6	0.98	15.3	2373	0.17	89.0
		365	bdl	0.20	2.78	283	154	6.01	0.34	2.16	94.2	0.01	950
		31,442	0.53	0.08	6.60	289	43,323	23.7	0.09	11.0	6145	0.01	194
		26,254	10.6	1.59	3.52	600	5316	347	0.27	3.40	6181	0.01	322
		72,069	11.0	0.21	3.22	470	12,479	60.1	0.12	2.24	1635	bdl	796
		Max	72,069	11.0	1.59	6.60	600	43,323	347	0.98	15.3	6181	0.17
	Min	365	bdl	bdl	2.78	283	154	6.01	0.09	2.16	94.2	bdl	89.0
	Avg	26,319	4.43	0.42	4.10	401	13,157	92.0	0.36	6.81	3286	0.04	470

Abbreviations: Ccp=Chalcopyrite. Notes: bdl, below detection limit; Non-detectable samples were assigned zero values in statistical computations.

Chaganbulagen deposit were obtained to be 207~393 °C, with an average of 331 °C; whereas the application of the FAS and FA6 geothermometer proposed by Zhang et al. (2022) were obtained as 218~333 °C (averaging 272 °C) and 208~336 °C (averaging 278 °C), respectively (Table 3). It can be seen that the temperatures estimated using the FA geothermometer are relatively lower than the GGIMFis geothermometer, but both estimation methods show that the sphalerite precipitation temperature decreases clearly from the early Cu-Zn stage to the late Ag stage. On balance, the temperatures calculated by the FAS and FA6 geothermometer are considered to be closer to the temperatures of fluid inclusions measured by Sun et al. (2024) (162–318 °C) and Li et al. (2016) (173–340 °C) in Chaganbulagen deposit.

From the InGa-InIn diagram (Fig. 9) (Zhang, 1987), the datapoints of SpI, SpII-a, SpII-b, and SpIII-a all fall in the I-Magmatic-hydrothermal region ($Ga/In < 1$), while most of the data points of the SpIII-b and SpIV in the late stage fall in the III-Sedimentary-reformed region ($Ga/In > 1$). Combined with previous studies of H-O-S isotopes in Chaganbulagen deposit (Li et al., 2016; Sun et al., 2024), it suggests that the genesis type of Chaganbulagen deposit is of the magmatic-hydrothermal type, whereas it may have been affected by atmospheric precipitation and mixed with materials from the strata in the late stage.

6.3. Enrichment process and spatial distribution of indium

The physicochemical conditions of the ore-forming fluid and the type and concentration of the metal ligand have an important influence on the distribution of indium in sphalerite. Temperature is the main factor that controls the occurrence of indium enrichment in the deposits, and

Table 3

The calculated temperature (°C) of sphalerite in Chaganbulagen deposit.

Mineralization stage	Sample		GGIMFis		FA geothermometer	
			PC1 *	T	T _{FAS}	T _{FA6}
Cu-Zn	SpI	Max	-2.57	393	333	336
		Min	-3.39	348	286	298
		Avg	-2.85	363	308	312
Cu-Pb-Zn	SpII-a	Max	-2.14	379	283	301
		Min	-3.15	324	267	275
		Avg	-2.50	344	274	285
	SpII-b	Max	-2.11	348	288	301
		Min	-2.57	323	265	279
		Avg	-2.37	337	275	286
Pb-Zn-Ag	SpIII-a	Max	-1.73	311	272	270
		Min	-1.90	302	257	257
		Avg	-1.83	308	266	264
	SpIII-b	Max	-1.62	341	295	292
		Min	-2.45	296	218	239
		Avg	-2.02	318	263	272
Ag	SpIV	Max	0.01	335	270	277
		Min	-2.34	207	224	208
		Avg	-1.25	276	248	247

Note: GGIMFis temperature is calculated using the formula proposed by Frenzel et al. (2016), FA geothermometer temperature is calculated using the Excel program from Zhang et al. (2021).

Liu et al., (1984) noted that in most cases, high temperature conditions are conducive to indium enrichment. Meanwhile, experimental studies have shown that indium mainly migrates as $InCl_4^-$ and $InClOH^+$ complex ions in the fluid of magmatic hydrothermal system, and the binding

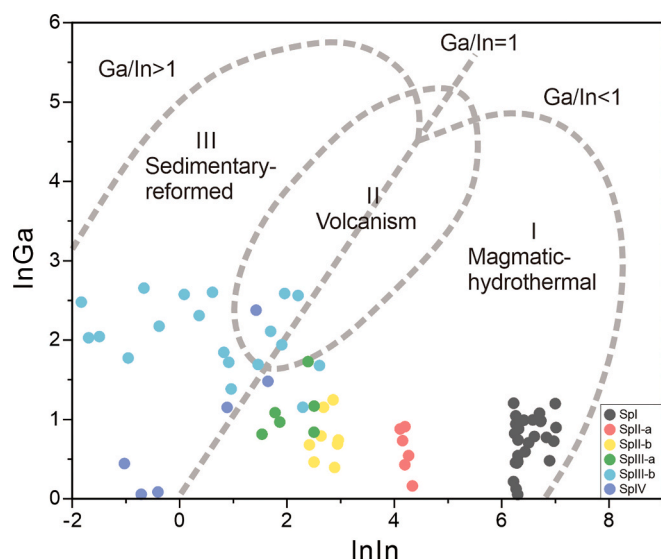


Fig. 9. LnGa-LnIn diagram of sphalerite (modified from Zhang, 1987).

ability of indium and complex ions depends on temperature change, the higher temperature, the more favorable, reaching the peak at 300–350 °C (Seward et al., 2000). Shimizu and Morishita (2012) found that indium is transported in high-temperature fluids (≥ 300 °C) by studying fluid inclusions in indium-bearing sphalerite in Japan Toyoha deposit. The temperature of fluid inclusions in roquesite-bearing tin ores from the Omodani and Akenobe deposits, also in Japan, is limited to 285–310 °C (Shimizu and Kato, 1991). In Chaganbulagen deposit, fluid inclusion homogeneous temperature and sphalerite geothermometer both show SpI temperatures of the early Cu-Zn stage greater than 300 °C. Sphalerite (SpI) and chalcopyrite (CcpI) from Cu-Zn stage containing high-In contents may related to the increased affinity of In for chlorine complex ions at high temperatures. Xiao et al. (2023) proposed that $\text{Cu}^+ + \text{In}^{3+} \rightarrow 2\text{Zn}^{2+}$ and $\text{Ag}^+ + \text{In}^{3+} \rightarrow 2\text{Zn}^{2+}$ are two essential substitution mechanisms for In element to enter sphalerite, and concluded through dynamical simulation that the suitable temperature for $\text{Cu}^+ + \text{In}^{3+} \rightarrow 2\text{Zn}^{2+}$ scheme is 300–325 °C, while that for $\text{Ag}^+ + \text{In}^{3+} \rightarrow 2\text{Zn}^{2+}$ is 225–250 °C. Concurrently, microscopic thermometry of fluid inclusions in Chaganbulagen deposit by previous works (Li et al., 2016) demonstrated that the early, middle, and late homogenization temperatures were 261–340 °C, 209–265 °C, and 173–219 °C, respectively, which matched with the appropriate temperatures for the two substitution mechanisms mentioned above. This indicates that the temperature may be the most important factor controlling the distribution of In in the sphalerite of Chaganbulagen deposit.

Except for temperature, the composition of the ore-forming fluid is also a key factor influencing indium enrichment. Indium usually enriched in Sn-rich deposits, whereas the concentration of indium in the fluids of Sn-poor deposits is usually much lower (Zhang et al., 2003, 2007; Zhu et al., 2006). Besides Sn, other metallic elements like Cu, Zn, Cd and Fe significantly affect indium accumulation (Cook et al., 2011; Shimizu and Morishita, 2012; Dill et al., 2013; Valkama et al., 2016). Cassiterite has been observed in Pb-Zn-Ag mineralization stage of Chaganbulagen deposit (Fig. 3h), while the high-Sn content in SpIII-b (averaging 120 ppm) and CcpIII (averaging 926 ppm) proves that the fluids in this stage are Sn-rich, but In is not enriched in this stage, suggesting that no potential relationship between In and Sn in Chaganbulagen deposit. This may be due to that the sphalerite precipitated at Pb-Zn-Ag stage was in an In-poor fluid environment and a large amount of In was consumed during precipitation in the earlier stages. Indium mineralization was primarily enriched during the early Cu-Zn stage (Fig. 10), with significant concentrations in both SpI (averaging 656 ppm) and CcpI (averaging 270 ppm). This indicates a close association

between indium enrichment and early-stage Cu-rich fluids, consistent with findings by Ye et al. (2023) in their study of the Inner Mongolia Dajing Sn-Cu polymetallic deposit. The “indium window” effect is not significant because Cd is relatively enriched in all stages of Chaganbulagen deposit (Fig. 5). Density functional theory (DFT) calculations show that the presence of Fe-impurities facilitates the entry of In into sphalerite crystals and that a positive correlation between Fe and In in sphalerite (Yang et al., 2022). The overall trend of In and Fe in sphalerite of Chaganbulagen deposit also show a positive correlation, and localized increases in Fe in SpIII-b may be due to additional atmospheric precipitation mixing Fe of the strata (Fig. 5).

With regard to the planar distribution, Cu-Zn mineralization stage of No. X orebody, which is closest to porphyry mineralization (1.5–3 km), sphalerite has the highest indium content, up to 1112 ppm; Cu-Pb-Zn and Pb-Zn-Ag mineralization stages of No. X, VIII, and II orebodies, which are 3.0–5.5 km away from the porphyry mineralization, sphalerite has a high indium content, up to 76.8 ppm; and Ag mineralization stage of No. I and V orebodies, which are the furthest away from the porphyry mineralization (5.5–8 km), sphalerite contains essentially no indium, only a few ppm. Consequently, the indium contents of the sphalerite in Chaganbulagen deposit shows a decreasing trend with increasing distance from the porphyry Mo (Cu) mineralization hydrothermal center, similar to the phenomenon observed in Baoshan Pb-Zn polymetallic deposit (Wu et al., 2023) and Dulong Pb-Zn deposit (Xu et al., 2021b). From the vertical distribution, the indium concentration in No. X orebody exhibits a progression of enhanced In enrichment with depth, where sphalerite (SpI) from the drill hole ZK15007 attains 1112 ppm In at 836-meter depth (averaging 653 ppm), contrasting with sphalerite (SpIII-a) from shallow-level of No. X orebody displaying depleted In values (averaging 10.4 ppm). This positive correlation between indium content and depth within the same deposit has also been found in Guangxi Dachang deposit (Pu, 2000; Dai et al., 2012) and Hunan Baoshan deposit (Wu et al., 2023). Indium in chalcopyrite follows a similar enrichment pattern as in sphalerite.

In summary, the No. X orebody of Chaganbulagen deposit hosts significant indium resources, with its mineralization exhibiting systematic spatial-temporal patterns intimately associated with the porphyry Mo (Cu) mineralization hydrothermal center. Detailed geochemical analyses reveal that indium enrichment in sphalerite and chalcopyrite demonstrates a progressive decline from early to late stages of fluid evolution (Fig. 10a, b), a trend that corresponds exactly to the deposit's established mineralization zoning and mineralization stages (Fig. 11). The regular spatial and temporal distribution patterns are closely related to temperature and fluid composition.

6.4. Implication for exploration

Elucidating the occurrence mechanisms and enrichment processes of indium not only enhances mineral extraction and processing efficiency, but more fundamentally, establishes theoretical basis for indium exploration (Xu et al., 2021a; Xu et al., 2021b; Chen et al., 2023). Sphalerite as a main host mineral for indium, and 95 % of the world's indium is currently recovered from sphalerite (Werner et al., 2017). Chalcopyrite has a low indium content (Serranti et al., 2002), especially in chalcopyrite and sphalerite symbiotic structures, where indium tends to preferentially occur in sphalerite (George et al., 2016; Xu and Li, 2018), but high indium content chalcopyrite also has been reported in some deposits. For example, in the Neves-Corvo Sn-Cu deposit in Portugal, chalcopyrite contains up to 1800×10^{-6} indium (Gaspar, 2002). In Chaganbulagen deposit, indium is not only enriched in early sphalerite (SpI), the early stage chalcopyrite (CcpI) also contains significant In (up to 544 ppm), so chalcopyrite should not be ignored when indium is extracted and utilized.

It's worth noting that Cd is enriched in all stages in Chaganbulagen deposit, and the content can be as high as 7742 ppm. Meanwhile, Cd in all paragenetic sphalerite stages attains the industrial grade for

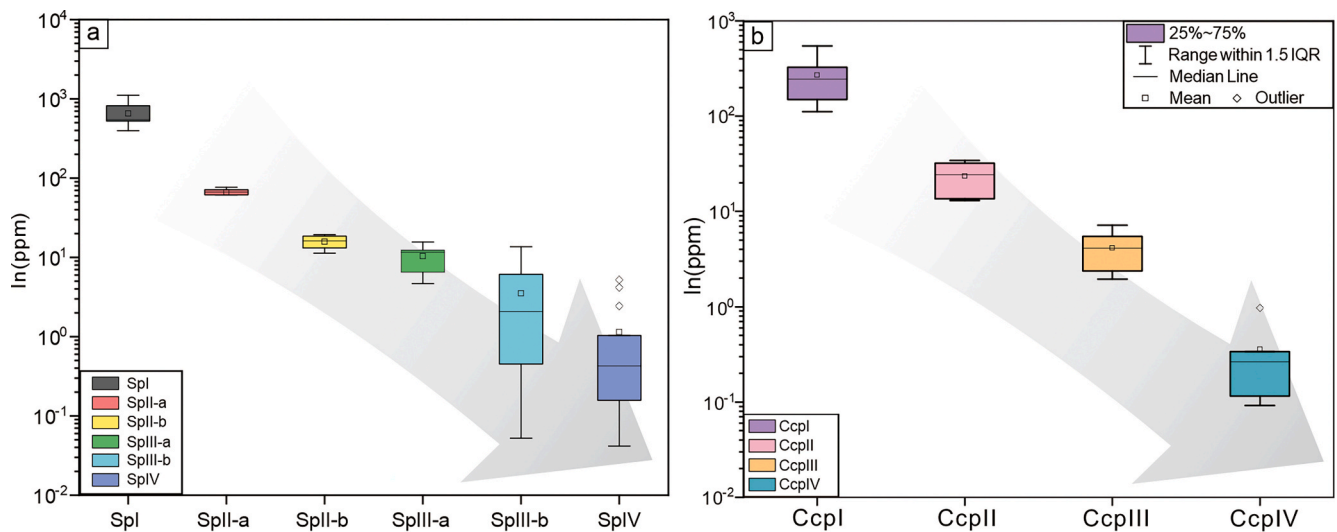


Fig. 10. Box-whisker plots showing In content in different generations of sphalerite (a) and chalcopyrite (b) of Chaganbulagen deposit.

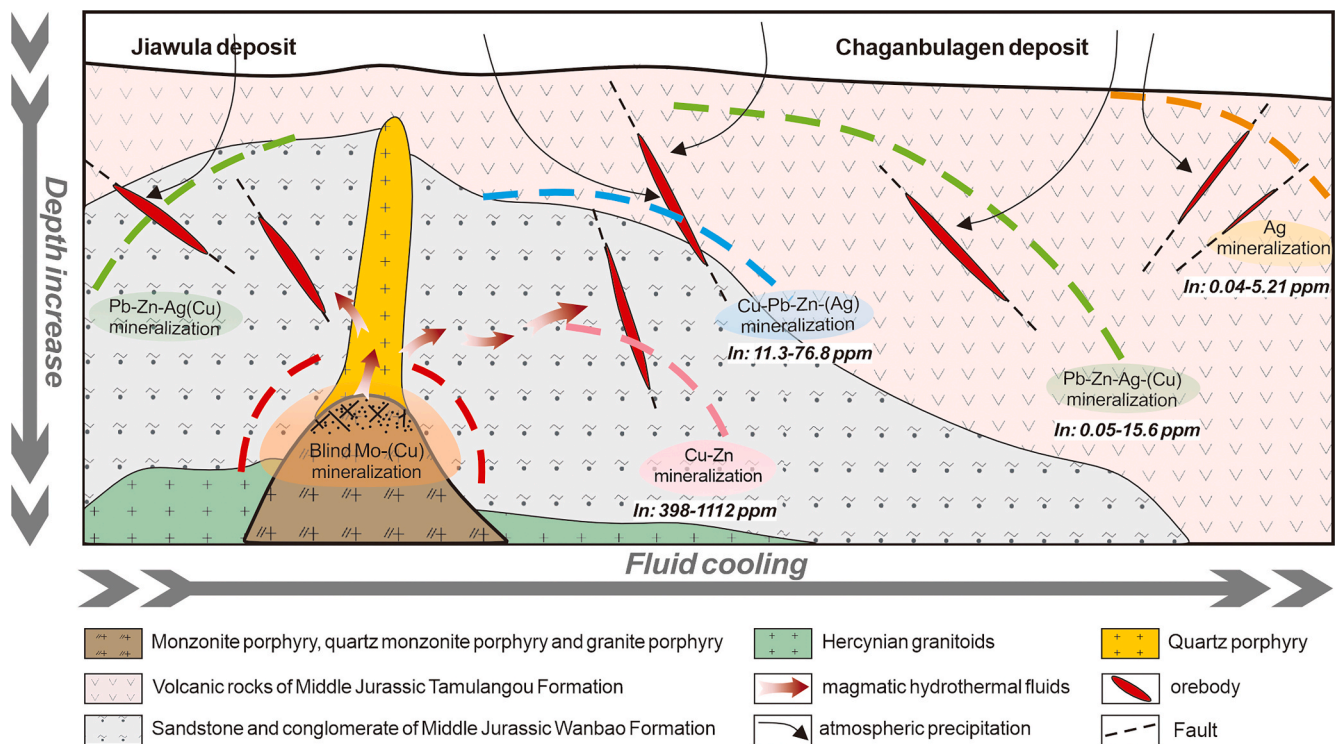


Fig. 11. Schematic metallogenic model of Chaganbulagen deposit.

comprehensive utilization. It is obviously higher than the general magmatic-hydrothermal type deposits: for example, the average Cd content in Yunnan Dulong deposit is 1625 ppm (Ye et al., 2017) and that in Henan Chitudian deposit is 3429 ppm (Tian et al., 2015), which shows that the Cd in Chaganbulagen deposits should not be ignored, and similarly, Cd in the medium- to high-temperature deposits associated with magmatic hydrothermal fluids should also be emphasized.

The indium mineralization in Chaganbulagen deposit is centered on porphyry Mo (Cu) mineralization hydrothermal center. Earlier stage, higher temperature and deeper spatially precipitated sphalerite and chalcopyrite have higher indium contents, generally 2–3 orders of magnitude higher than those precipitated at later stage. This enrichment pattern shows remarkable similarity to the spatial variation of indium observed in Hunan Baoshan deposit (Wu et al., 2023). On this basis, in

combination with the spatial location of the indium-rich sphalerite and chalcopyrite in this study, proposed that the Jia-Cha ore field, centered on the porphyry Mo (Cu) mineralization within 3 km to the northwest and southeast, and the deeper parts of the field are favorable areas for future indium ore exploration (Fig. 11).

7. Conclusions

- (1) The Chaganbulagen deposit can be divided into four stages of mineralization, with six types of sphalerite and four generations of chalcopyrite identified. The critical metals In and Cd are predominantly hosted in sphalerite and chalcopyrite by isomorphisms.

- (2) In mainly enriched in Cu-Zn mineralization stage, In contents in sphalerite and chalcopyrite of Chaganbulagen deposit shows a gradual decrease from early to late stages with fluid evolution. The temperature and composition of the ore-forming fluid is a favorable condition for controlling the distribution of indium in sphalerite of Chaganbulagen deposit. The regular spatial and temporal distribution patterns are closely related to temperature and fluid composition.
- (3) The Jia-Cha ore field, centered on the porphyry Mo (Cu) mineralization hydrothermal center within 3 km to the northwest and southeast, and the deeper parts of the field are favorable areas for future indium ore exploration.

CRedit authorship contribution statement

Mingyu Cao: Writing – original draft, Investigation, Conceptualization. **Jiaqi Xu:** Data curation. **Huimin Su:** Writing – review & editing, Methodology, Conceptualization. **Yushan Zuo:** Resources, Project administration, Conceptualization. **Wentian Mi:** Writing – review & editing, Resources, Funding acquisition. **Shanming Zhang:** Validation, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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